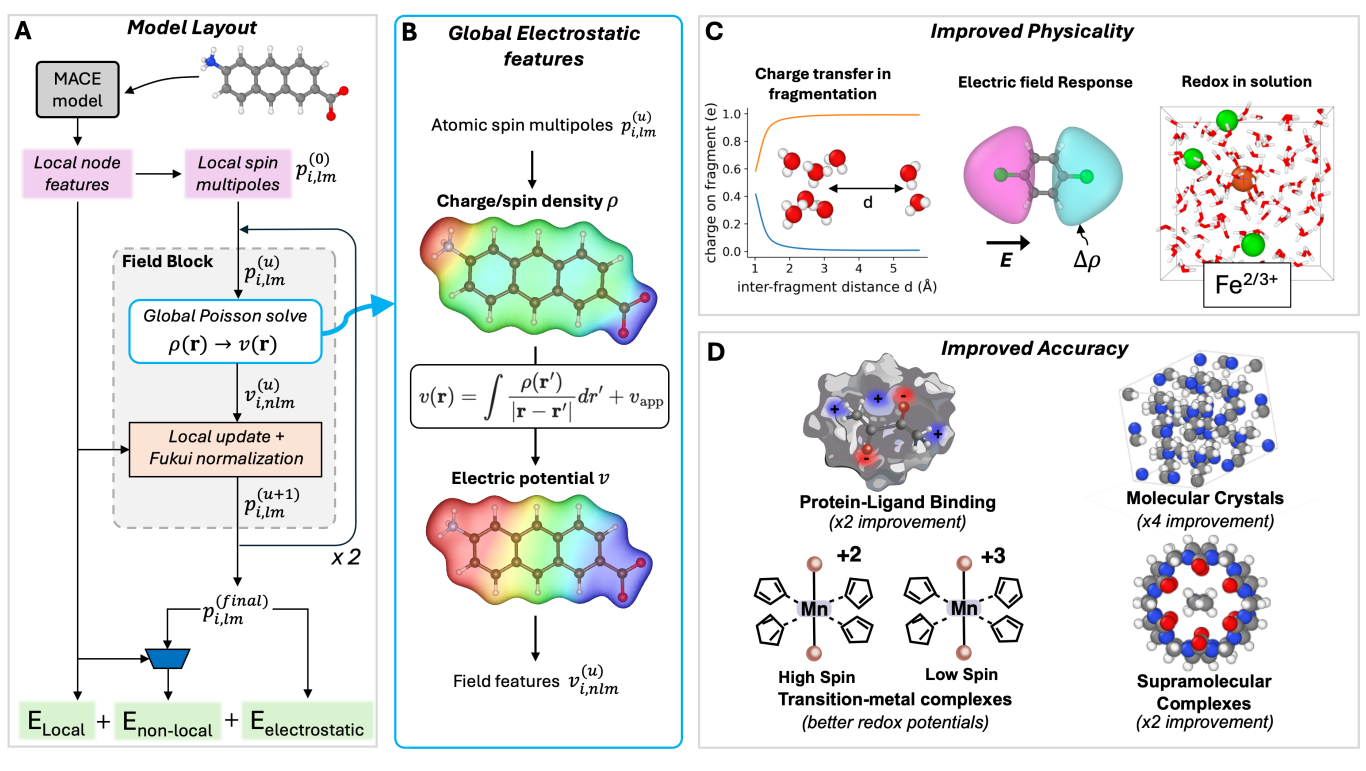

A
Model Layout
MACE model
Local node features
Local spin multipoles
$p^{(0)}_{i,lm}$
Field Block
$p^{(u)}_{i,lm}$
Global Poisson solve
$\rho(\mathbf{r}) \rightarrow v(\mathbf{r})$
$v^{(u)}_{i,nlm}$
Local update + Fukui normalization
$p^{(u+1)}_{i,lm}$
x 2
$p^{(final)}_{i,lm}$
E_{Local} + $E_{non\text{-}local}$ + $E_{electrostatic}$
B
Global Electrostatic features
Atomic spin multipoles $p^{(u)}_{i,lm}$
Charge/spin density ρ
$v(\mathbf{r}) = \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dr' + v_{\text{app}}$
Electric potential v
Field features $v^{(u)}_{i,nlm}$
C
Improved Physicality
Charge transfer in fragmentation
charge on fragment (e)
1.0
0.8
0.6
0.4
0.2
0.0
d
1 2 3 4 5
inter-fragment distance d (Å)
Electric field Response
E
Δρ
Redox in solution
Fe2/3+
D
Improved Accuracy
Protein-Ligand Binding
(x2 improvement)
Molecular Crystals
(x4 improvement)
+2
Mn
+3
Mn
High Spin
Low Spin
Transition-metal complexes
(better redox potentials)
Supramolecular Complexes
(x2 improvement)